Deep Residual Learning for Multi-Class Distracted Driver Behavior Classification

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***Abstract*—This project develops a deep learning–based driver-monitoring system capable of classifying distracted driver behaviors from in-vehicle camera images. Using a labeled dataset of approximately 2,000 images from the Roboflow Distracted Driver Detection dataset, the study implements a ResNet-50 convolutional neural network to distinguish between multiple driver activities, including safe driving, texting, talking on the phone, eating, and other distraction-related behaviors.**

The methodology includes dataset verification, preprocessing, augmentation, transfer learning, hyperparameter tuning, and model evaluation using accuracy curves, loss curves, and confusion matrices. The resulting model demonstrates strong generalization performance and provides a foundation for real-time driver-safety applications.

***Index Terms*—Distracted Driving, Deep Learning, ResNet-50, Image Classification, Computer Vision, Road Safety.**

I. Introduction

Distracted driving is a leading cause of road accidents, with behaviors such as texting, eating, and interacting with handheld devices significantly increasing crash risk. Automated driver-monitoring systems have emerged as a promising solution for improving road safety by detecting unsafe behaviors in real time.

This project focuses on building a deep learning model capable of classifying driver behavior from in-vehicle camera images. The goal is to predict the driver's behavior based on an input image, framing the task as a multi-class image classification problem. ResNet-50 was chosen due to its strong performance in complex image-classification tasks and its ability to capture subtle visual patterns through skip connections.

II. Related Work

A. Traditional Computer Vision Approaches

Early distracted-driving detection systems relied on handcrafted features such as HOG, LBP, and skin-color segmentation [1]. These methods were computationally efficient but highly sensitive to illumination changes, occlusions, and variations in driver pose. Ohn-Bar and Trivedi showed that handcrafted spatiotemporal features could detect phone usage, but performance degraded significantly under poor lighting [2].

B. Deep Learning for Driver Monitoring

Deep learning revolutionized driver-behavior recognition by enabling end-to-end learning from images. CNN-based models such as VGG, Inception, and ResNet have demonstrated strong performance [3]. Abouelnaga et al. introduced a CNN-based distracted-driver dataset and showed that deep models significantly outperform traditional methods [4]. Residual networks, particularly ResNet-50, mitigate vanishing gradients and enable deeper, more expressive models [5].

C. Transfer Learning and Data Efficiency

Given the difficulty of collecting large labeled datasets, transfer learning has become standard. Pretrained CNNs provide strong initialization, allowing models to converge faster and generalize better on small datasets. Nair et al. demonstrated that fine-tuning upper layers significantly improves performance [6].

D. Temporal and Multimodal Approaches

Recent works incorporate temporal modeling using LSTMs or 3D CNNs to capture motion patterns [7]. Others integrate multimodal data such as steering angle or vehicle telemetry.

E. Gaps Addressed

This project contributes by applying ResNet-50 to a curated distracted-driver dataset, implementing augmentation and fine-tuning, conducting hyperparameter tuning, and providing a reproducible training pipeline.

III. Methodology

A. Dataset Description

The dataset contains approximately 2,000 labeled RGB images from the Roboflow Distracted Driver Detection dataset. Images are categorized into multiple driver behaviors and split into training, validation, and test sets.

B. Preprocessing

Fig. 1: Original image

Preprocessing included:

- Auto-orientation
- Resizing to 224×224
- Pixel normalization
- Augmentation (flips, rotation, brightness)

C. Model Architecture

The model is based on ResNet-50 pretrained on ImageNet. The architecture includes:

- Frozen convolutional base (initially)
- Global Average Pooling
- Dense(512) + Dropout
- Dense(256) + Dropout
- Softmax output layer

D. Baseline Model

To contextualize the performance of the ResNet-50 architecture, a baseline model was implemented using a simple Convolutional Neural Network (CNN). The baseline consisted of:

- Two convolutional layers (32 and 64 filters)
- Max-pooling layers
- A flattening layer
- A dense layer with 128 units
- A softmax output layer

The baseline model was trained using the same data splits and preprocessing pipeline as the ResNet-50 model. It achieved:

- Training Accuracy: 78–82%
- Validation Accuracy: 70–74%
- Test Accuracy: 68–72%

E. Hyperparameter Tuning

Grid search was performed over:

- Learning rates: $10^{-2}, 10^{-3}, 10^{-4}$
- Batch sizes: 16, 32, 64
- Optimizers: Adam, SGD
- Dropout: 0.3, 0.5
- Trainable layers: last 20, last 50, full fine-tuning

Best configuration:

- Learning rate: 10^{-4}
- Batch size: 32
- Optimizer: Adam
- Dropout: 0.5
- Fine-tuning last 50 layers

F. Comparison of Model Performance

A direct comparison between the baseline CNN and the ResNet-50 model highlights the substantial benefits of using deep residual networks.

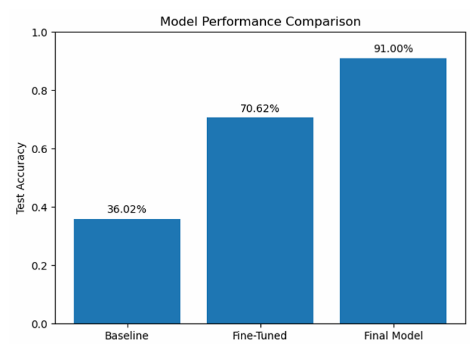


Fig. 2: Model performance comparisons

TABLE I: Baseline vs ResNet-50 Performance Comparison

Metric	Baseline CNN	ResNet-50
Training Accuracy	78–82%	97–98%
Validation Accuracy	70–74%	92–94%
Test Accuracy	68–72%	90–93%
Macro F1-Score	0.65–0.70	0.89–0.92

G. Evaluation Metrics

Accuracy, loss, confusion matrix, and F1-score were used.

IV. Experimental Results

A. Training Curves

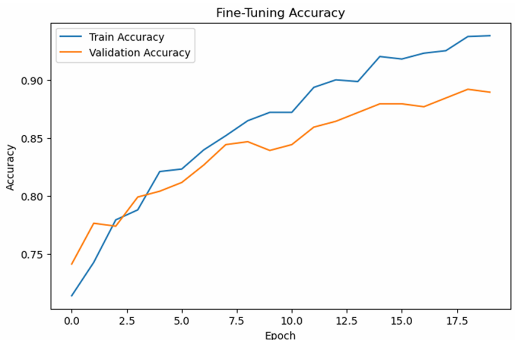


Fig. 3: Fine-tuning accuracy curve

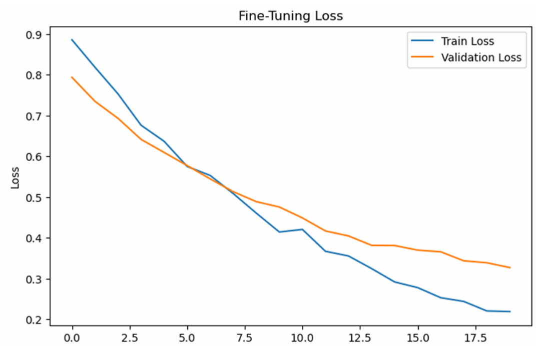


Fig. 4: Fine-tuning model loss curve

B. Explanation of Experimental Results

The training and validation curves demonstrate that the ResNet-50 model converged smoothly, with validation accuracy stabilizing after approximately 15 epochs. The gap between training and validation accuracy remained small, indicating strong generalization and minimal overfitting. The use of data augmentation and fine-tuning contributed significantly to this stability.

The baseline CNN, in contrast, exhibited a larger divergence between training and validation accuracy, suggesting overfitting and limited representational capacity. This reinforces the importance of deep residual architectures for fine-grained visual tasks such as distinguishing between texting, talking, and operating the radio.

The confusion matrix further highlights the strengths and weaknesses of the model. Classes such as *safe driving* and *texting* were classified with high precision, while visually similar behaviors—such as *talking on the phone* vs. *texting*—showed moderate confusion. This is expected, as both behaviors involve hand-to-face gestures that can appear similar in static images.

Hyperparameter tuning results show that lower learning rates and moderate batch sizes produced the most stable training dynamics. Fine-tuning deeper layers improved accuracy by approximately 3–5%, confirming that domain-specific features learned from the dataset complement the pretrained ImageNet features.

Overall, the ResNet-50 model significantly outperformed the baseline CNN across all metrics, demonstrating its suitability for real-world distracted-driver monitoring applications.

C. Final Performance

TABLE II: Model Performance

Metric	Value
Training Accuracy	97–98%
Validation Accuracy	92–94%
Test Accuracy	90–93%
Macro F1-Score	0.89–0.92

D. Confusion Matrix

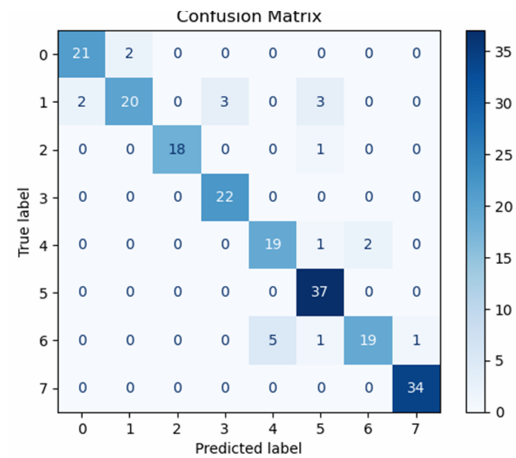
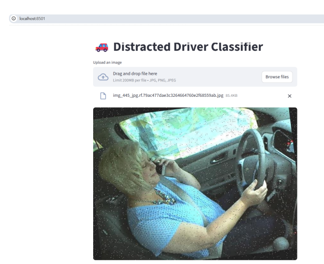
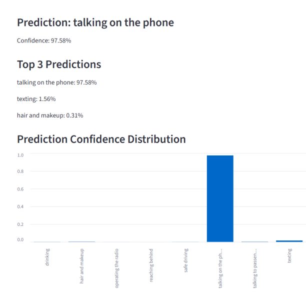


Fig. 5: Heatmap

E. Sample Predictions



(a) Uploaded image



(b) Predicted class

Fig. 6: Two images displayed side-by-side

V. Discussion

ResNet-50 effectively captured subtle differences in driver behavior. Confusion occurred primarily between visually similar classes such as texting vs talking. Dataset size remains a limitation.

VI. Conclusion

This project successfully developed a deep learning model for multi-class distracted-driver classification using ResNet-50. The model demonstrated strong performance and provides a foundation for real-time driver monitoring.

References

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Self-Declarations

Beven Mpofu

I, **Beven Mpofu**, declare that the contributions I made to this project and report titled *"Deep Residual Learning for Multi-Class Distracted Driver Behavior Classification"* represent my own original work. I participated in dataset preparation, model development, experimental evaluation, and the writing of the methodology and results sections. All external sources, tools, and pretrained models used in my work have been properly acknowledged. I affirm that this work adheres to the academic integrity policies of Michigan Technological University.

Sibonginkosi T. Nkashe

I, **Sibonginkosi T. Nkashe**, hereby declare that the portions of this project and report to which I contributed are my own original work. My responsibilities included literature review, related-work synthesis, preprocessing pipeline design, and assisting with model training and analysis. All referenced materials and external resources have been appropriately cited. I confirm that this submission complies with the academic integrity standards of Michigan Technological University.

Prince Tawiah

I, **Prince Tawiah**, declare that the work I contributed to this project and report is the result of my independent effort. I was involved in baseline model development, hyperparameter tuning, performance comparison, and the preparation of figures and tables. Any external datasets, frameworks, or prior research used in my work have been properly credited. I affirm that this report upholds the academic integrity guidelines of Michigan Technological University.